

The Brittleness of Plasticity: A Synthesis for Neural Network Stability via Mask Evolution Operators

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September 2025

Abstract

We study the stability–plasticity trade-off in continual learning through *Mask Evolution Operators* (MEOs): task-agnostic transformations applied at activation level that exert a restoring force on drifting representations. We frame MEO as a family of evolution operators over internal states and instantiate the *Identity Evolution* case as a stringent stress test: representations are lightly regularized toward a reference manifold while the model continues to learn.

We report smoke-test experiments on CIFAR-100 (10 tasks $\times$ 10 classes) using ResNet-50. On an Apple M1 (MPS backend), a full finetune baseline yields final average accuracy **6.28%**; an EWC baseline (best λ from a fast grid, then confirmed at 20 epochs) yields **6.21%**. These results are limited by short training schedules and CPU-class hardware; they serve as *sanity checks* validating our pipeline and metrics prior to full GPU runs. Code and configs are available at github.com/PaoloTCS/meo-continual-learning. We position MEO as a general operator framework; extending beyond Identity to controlled-evolution masks is left for future work.

Availability and Artifacts. Code, configs, and scripts: github.com/PaoloTCS/meo-continual-learning. Figures are generated from the repository’s `results/` directory.

1 Introduction

Catastrophic forgetting remains a central obstacle in continual learning: as models adapt to new tasks, internal representations drift, eroding performance on earlier tasks. Classical mitigations constrain weights (EWC, SI) or expand capacity (dynamic architectures), but each carries trade-offs in flexibility, compute, or memory.

We propose a representation-centric view: *Mask Evolution Operators* (MEOs) act directly on activations, introducing a tunable, vector-field-like restoring force that discourages destructive drift while preserving useful plasticity. Conceptually, MEO decouples *where* learning occurs (the usual optimizer over weights) from *how* state evolves in latent space (a mask-based operator). Practically, it is lightweight and architecture-agnostic.

This paper contributes: (i) a formalization of MEO as an evolution operator over activations; (ii) an *Identity Evolution* instantiation used as a *stress test* to probe stability without freezing learning; (iii) a reproducible pipeline with drift metrics; (iv) smoke-test empirical results that validate the setup and motivate full GPU experiments.

2 Related Work

Weight-space constraints. EWC penalizes deviations from Fisher-weighted optima [?]. Synaptic Intelligence and related approaches follow similar principles with different estimators. **Representation-space control.** Feature-level regularizers, contrastive consolidation, and orthogonality constraints modulate latent geometry directly. MEO generalizes this idea by treating the controller as an explicit evolution operator. **Continual learning protocols.** Split CIFAR-100 and similar benchmarks remain standard for assessing drift and forgetting under multi-task streams.

3 Mask Evolution Operators

Let $h \in \mathbb{R}^d$ denote an activation vector at some layer. A *Mask Evolution Operator* is a map

$$\mathcal{M}_\theta : \mathbb{R}^d \rightarrow \mathbb{R}^d, \quad \tilde{h} = \mathcal{M}_\theta(h), \quad (1)$$

parameterized by θ (which may be learned or fixed), applied during the forward pass. We define a *mask loss* that penalizes deviations from a reference state $h^\star$:

$$\mathcal{L}_{\text{mask}}(h; h^\star) = \alpha \|\mathcal{M}_\theta(h) - h^\star\|_2^2, \quad (2)$$

with stiffness $\alpha > 0$ governing the restoring magnitude. The total training loss is

$$\mathcal{L} = \mathcal{L}_{\text{task}} + \mathcal{L}_{\text{mask}}. \quad (3)$$

Identity Evolution as stress test. In this paper we instantiate $\mathcal{M}_\theta$ as the identity Id, anchoring representations to a fixed $h^\star$ captured early in training (Task 1). This deliberately favors stability and tests whether a light restoring force can deter destructive drift while still allowing useful adaptation. In future work we will allow $\mathcal{M}_\theta$ to evolve (e.g., low-rank adaptations, geodesic updates), enabling controlled motion of $h^\star$ across tasks.

Drift metric. We measure representation drift at layer ℓ via

$$\Delta_\ell = \mathbb{E}_{x \sim \mathcal{D}} \left[\|h_\ell(x; t) - h_\ell^\star(x)\|_2 \right], \quad (4)$$

comparing current activations to the reference. We report per-epoch and per-task aggregates.

4 Baselines and Training

Finetune. Standard ERM without rehearsal or consolidation.

EWC. Online EWC with Fisher estimated by gradient-squared averages over minibatches; we sweep λ on a log scale and confirm the best at longer training.

Architecture and data. ResNet-50 on CIFAR-100 split into 10 tasks of 10 classes each; shared head with 100-way classification. Input preprocessing follows standard CIFAR normalization.

Hardware and schedule (smoke tests). All runs were executed on an Apple M1 (macOS, PyTorch MPS backend). For feasibility we used short schedules: 10 epochs per task for the grid, then 20 for the best EWC λ . These are *sanity checks* to validate the pipeline; full GPU results will follow.

Optimization and Fisher details. SGD with momentum 0.9, weight decay 5×10^{-4} , cosine LR, batch size 128 (reduced if memory-bound). For EWC, Fisher accumulation uses batch-averaged g^2 with online decay $\gamma = 0.9$; we normalized by sample count to stabilize scale.

5 Results

Table ?? summarizes final average accuracies (mean over the 10 tasks after the full stream). Numbers reflect the smoke-test setting on M1/MPS and should be interpreted as preliminary.

Method	Final Avg. Acc. (%)	Notes
Finetune	6.28	10 tasks $\times$ 10 epochs
EWC (best $\lambda=30$)	6.21	grid@10 epochs, confirm@20

Table 1: Smoke-test results on CIFAR-100 (ResNet-50) using Apple M1 (MPS). Values are expectedly low under short schedules and CPU-class hardware; they validate the pipeline prior to full GPU runs.

Drift and stiffness. Figure ?? shows per-epoch drift under a short schedule; Figure ?? sweeps the stiffness α for MEO. Both illustrate expected trends but should be considered qualitative under these constraints.

6 Discussion

Identity as stress test. A key critique of operator rigidity is addressed by positioning the Identity Evolution as a *stress test*: it intentionally biases toward stability to probe whether even a simple operator can resist destructive drift without prohibiting adaptation. Our smoke tests indicate the pipeline works as intended; definitive conclusions require full GPU runs and non-identity operators.

Beyond Identity. The MEO formalism naturally admits controlled-evolution operators (e.g., low-rank masks, proximal/geodesic updates) that move h^* along a constrained manifold. We hypothesize these will better preserve *beneficial* plasticity while suppressing destructive drift.

Limitations. Short schedules, MPS backend, and absence of rehearsal constrain absolute performance. Our focus here is methodological clarity and reproducibility; scaling up is orthogonal and planned.

7 Reproducibility

Repository: github.com/PaoloTCS/meo-continual-learning. We provide configs, scripts for EWC grid sweeping and paper auto-update, and plotting utilities for drift and α -sweeps. The exact runs reported here used seed 42; see **results/** JSON logs.

Acknowledgements

This research was conducted in a hybrid human–AI collaborative workflow. The author served as scientific lead, with **ChatGPT (GPT-5)** acting as the primary technical assistant and editor. Additional contributions came from **Claude**, **Grok**, and **Gemini**, which provided peer-review style

Algorithm 1 Training with MEO (Identity Evolution stress test)

Require: tasks $\{\mathcal{T}_k\}_{k=1}^K$, model f_ϕ , reference activations h^* (from Task 1), stiffness α

- 1: **for** $k = 1$ to K **do**
- 2: **for** minibatch $(x, y) \sim \mathcal{T}_k$ **do**
- 3: $h \leftarrow \text{forward_activations}(f_\phi, x)$
- 4: $\tilde{h} \leftarrow h$ ▷ Identity evolution
- 5: $\mathcal{L}_{\text{task}} \leftarrow \text{CE}(f_\phi(x), y)$
- 6: $\mathcal{L}_{\text{mask}} \leftarrow \alpha \|\tilde{h} - h^*\|_2^2$
- 7: $\mathcal{L} \leftarrow \mathcal{L}_{\text{task}} + \mathcal{L}_{\text{mask}}$
- 8: Update ϕ by SGD
- 9: **end for**
- 10: Optionally update statistics (drift, Fisher, etc.)
- 11: **end for**

feedback and alternative perspectives. This iterative, dialogic process reflects our broader R&D philosophy: scientific progress as a conversational, multi-agent activity in which human insight and machine reasoning co-evolve.

References

- [1] Kirkpatrick, J., et al. Overcoming catastrophic forgetting in neural networks. *PNAS*, 2017.



Figure 1: **Drift per epoch (smoke test).** CIFAR-100, ResNet-50, Apple M1 (MPS), seed=42, short schedule. Higher-than-ideal magnitudes reflect limited training; the trend illustrates how the restoring term affects representation movement. Replace this placeholder with the generated figure when available.



Figure 2: **Stiffness sweep α (smoke test)**. Validation of the MEO stiffness parameter under short schedules. Mark the operating point used in the main experiment. Replace this placeholder with the generated sweep figure when available.